GREAT LAKES INSTITUTE of MANAGEMENT GURGAON	**Postgraduate Diploma in Management (PGDM): 2024-26** **Term 5 | End-Term Examination** **Deep Learning & Natural Language Processing**	Max Marks: 40 Duration: 2 Hrs No. of Page(s): 3

Name:	Roll No.:

Instructions

- Write the correct question number before answering each question.
- Internet access is strictly prohibited during the exam.
- Only a laptop with Anaconda and Jupyter Notebook open is allowed. No other applications or tools should be used.
- Submit your solution only in Jupyter Notebook format **(**`.ipynb`**)**. Submissions in any other format (e.g., **`.py`**, **`.zip`**, **`.rar`**) will not be evaluated.
- Do not write your name or roll number anywhere in the notebook file. Instead, write the 9-digit code provided by the examiner during the examination clearly at the beginning of your notebook.
- Ensure that the output of each question is visible in the notebook (i.e., all cells must be executed before submission).
- Answer only what has been asked in the question. Any extra, irrelevant, or unasked content will lead to negative marking.
- Ensure that your notebook runs from start to end without errors before submission.
- All interpretations, explanations, and theoretical answers must be written in Markdown cells (not as code comments).

Answer all questions

Problem statement: A leading bank wants to improve its loan approval process using data-driven methods. They have collected data on 614 past loan applications, including applicant demographics, income details, employment status, credit history, and property type. The goal is to predict whether a loan should be approved (Y) or rejected (N) based on these features.

Dataset Information

The dataset(data.csv) includes the following features:

Loan_ID	Unique identifier for each loan application.
Gender	Gender of the applicant (Male / Female).
Married	Marital status of the applicant (Yes / No).
Dependents	Number of dependents (0, 1, 2, 3+).
Education	Education level of the applicant (Graduate / Not Graduate).
Self_Employed	Whether the applicant is self-employed (Yes / No).
ApplicantIncome	Monthly income of the applicant.
CoapplicantIncome	Monthly income of the co-applicant (if any).
LoanAmount	Loan amount requested.
Loan_Amount_Term	Term of the loan (in months).

Credit_History Indicates whether the applicant's credit history meets guidelines (1.0 = Yes, 0.0 = No).
Property_Area Area of the property (Urban, Semiurban, Rural).
Loan_Status Target variable indicating loan approval (Y = Approved, N = Not Approved).

1. Design and evaluate a Deep Neural Network for loan prediction with the following conditions:
   a. Use ReLU activation in all hidden layers. Report and interpret the model's performance on both the training and test sets. **[5 Marks, CO1/PO1]**
   b. Reconfigure the network by changing at least two elements of its architecture. Compare its performance on both the training and test sets. **[5 Marks, CO1/PO1]**

2. Apply and evaluate overfitting prevention techniques in your model with the following conditions:
   a. Implement at least three overfitting-prevention techniques. Evaluate how these techniques influence model performance by comparing the results with a baseline model without overfitting controls. **[5 Marks, CO2/PO2]**
   b. Explain and compare the underlying principles of dropout, early stopping, and L1/L2 regularization in the context of deep neural networks. **[5 Marks, CO2/PO2]**

3. From the context of core NLP and pretrained models, do the following:
   a. Explain the importance of text preprocessing in NLP. Discuss all steps in brief and it influences the quality of textual features. **[5 Marks, CO3/PO2]**
   b. Compare Bag-of-Words and TF-IDF as feature-representation techniques. Explain how each method captures information from text and discuss their strengths and limitations in text-classification tasks. **[5 Marks, CO3/PO2]**
   c. Describe the key components of transformer architectures, focusing on self-attention and contextual embeddings. Explain why these mechanisms improve performance in NLP tasks. **[5 Marks, CO3/PO2]**
   d. Compare transformer-based models (e.g., BERT, DistilBERT) with classical NLP approaches. Discuss their advantages, limitations, and scenarios where transformers may not outperform traditional methods. **[5 Marks, CO3/PO2]**